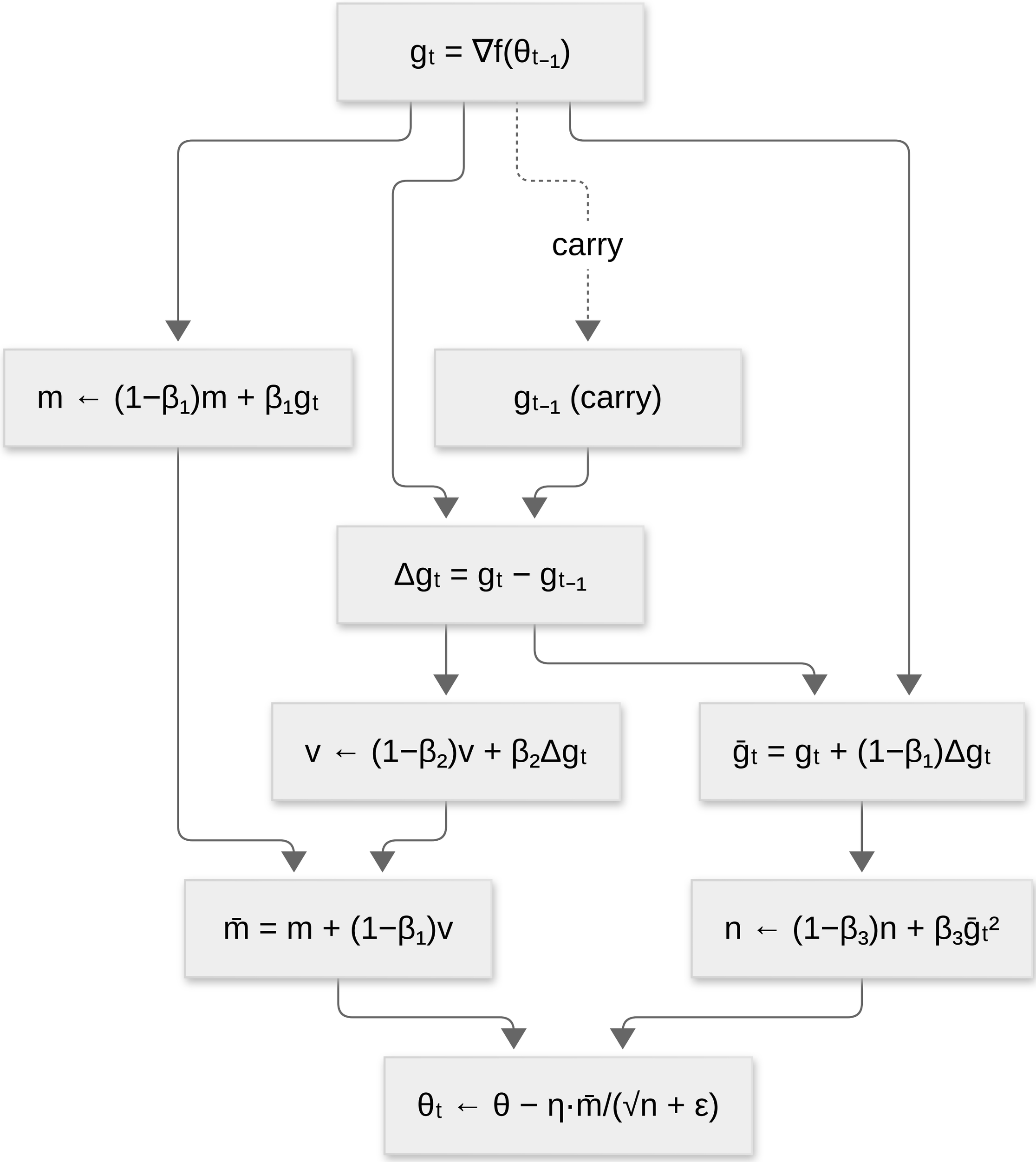

$g_t = \nabla f(\theta_{t-1})$
carry
$m \leftarrow (1-\beta_1)m + \beta_1 g_t$
g_{t-1} (carry)
$\Delta g_t = g_t - g_{t-1}$
$v \leftarrow (1-\beta_2)v + \beta_2 \Delta g_t$
$\bar{g}_t = g_t + (1-\beta_1)\Delta g_t$
$\bar{m} = m + (1-\beta_1)v$
$n \leftarrow (1-\beta_3)n + \beta_3 \bar{g}_t^2$
$\theta_t \leftarrow \theta - \eta \cdot \bar{m}/(\sqrt{n} + \varepsilon)$